



Module: Electronics, system components

Corrected Series of TD n° 02

Exercise 1:

1)

Generation	Characterized electronic component
First generation	- Use of vacuum tubes
Second generation	- Use of transistors
Third generation	- Use of integrated circuits
Fourth generation	- Use of very large-scale integrated circuits called VLSI (microprocessor)

2)

Generation	Characteristics of each Generation	Example
First generation	- Bigger, slower and less efficient - Use of punched cards - Not marketed.	- IBM UNIVAC I
Second generation	- Faster and smaller than the previous generation - Work with high-level languages.	- IBM 1620
Third generation	- Flexible and smaller - Well suited for commercial applications - Known as mini-computers.	- IBM 360 - PDP 8 - PDP 11
Fourth generation	- General purpose interactive machines - Enable rapid application development - Easier use for personal applications, - Can be easily used over a network.	-IBM PC -Apple II -Supercomputers like CRAYseries computers

Exercise 2:

1) Computers can generally be classified by size, power and areas of application

- 2)
- **Supercomputers:** Very large and very powerful computers (computing speed) which use tens of thousands of microprocessors to carry out certain tasks
 - **Embedded computers:** Is a component incorporated into a system (car, plane, washing machine, refrigerator, camera, etc.). These computers are built to perform a limited number of tasks (such as guidance or security).
 - **Microcomputers:** Represents the most common type of computer on the market. Their power continues to increase. There are different types of microcomputers: desktop computers, portable computers (called also notepads, notebook, personal computer, laptop), pocket computers (PDA, personal digital assistant), Tablet PC and body computers.
- 3) a comparative table between the five types of microcomputers studied according to power, autonomy, price, volume and portability.

Microcomputer types	Power	Volume	Portability
Desktop computers	Superior processing power	Bulky and heavy	Stationary
Portable computers (notepads)	Decent processing power, though gaming laptops have impressive processors	Compact and lightweight	Portable
Pocket computers (PDA, personal digital assistant),			
Tablet PC			
Body computers			

Exercise 3:

- 1) What does the acronym ENIAC mean?
(b) Electronic Numerical Integrator and Computer
- 2) ENIAC was:
(a) Electronic computer
- 3) Which of the following components are large, slow, and suffer from heat and maintenance issues?
(b) Vacuum tubes
- 4) Which of the following technologies was used by the third generation of computers?
(c) Integrated circuits
- 5) Which of the following computers are expensive and can process billions of instructions per second?
(a) Supercomputers
- 6) What do you call a computer that works with binary digits 0 and 1?
(a) Digital
- 7) Which of the following is called a handheld computer?
(b) Personal Digital Assistants (PDAs)
- 8) What are the essential components of a computer?
(a) A microprocessor
- 9) Which device is used as a standard pointing device in a graphical user environment?
(b) Mouse
- 10) Computer software can be defined as
(b) Instructions that tell the computer what to do

Exercise 4:

A keyboard is an essential input device for the computer, which allows commands to be sent to the computer:

- 1) What are the function keys?
The function keys are used to perform specific tasks. They are labeled as F1, F2, F3, and so on, up to F12. The functionality of these keys differs from program to program.
- 2) What are the delete keys?
The delete key is a key on most computer keyboards which typically is used to Delete key deletes a character next to the cursor
- 3) What are the input keys?
Enter keys used to move the cursor to the beginning of the next line. In a dialog box, press Enter to select the highlighted button
- 4) **Tab keys** used to move the cursor several spaces forward. You can also press Tab to move to the next text box on a form
- 5)
 - **1. Navigation keys.** These keys are used for moving around in documents or webpages and editing text. They include the arrow keys, Home, End, Page Up, Page Down, Delete, and Insert.
 - **2. Esc Key.** Used to Cancel the current task
 - **3. Numeric keypad.** The numeric keypad is handy for entering numbers quickly. The keys are grouped together in a block like a conventional calculator or adding machine.
 - **4. Control keys.** These keys are used alone or in combination with other keys to perform certain actions. The most frequently used control keys are Ctrl, Alt, the Windows logo key, and Esc.

Windows logo key 

Open the Start menu

Another key:

- **Typing (alphanumeric) keys.** These keys include the same letter, number, punctuation, and symbol keys found on a traditional typewriter.

Exercise 5:

- 1) tells the computer how to use its components
(a) Operating system
- 2) The physical parts that make up your computer are called
(c) Hardware
- 3) Computers use number systems to store data and perform calculations
(a) Binary
- 4).....is an electronic device that processes data and converts it into information
(a) Computer
- 5)is processed by the computer to extract the information
(b) Data
- 6)of a system includes the instructions or programs
(d) Software
- 7) Which of the following statements is true?
(a) The minicomputer runs faster than the microcomputer
- 8) Which of the following terms is not an operating system?
(c)Microsoft office xp
- 9) Give the term intruder
(a) Internet
- 10) The flashing symbol on the computer screen is called:
(b) Cursor

Good luck
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